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CSCE 580 Final

Question 1a

- a) Write the name of the paper and student presenter you chose.
 - a. Graph Neural Prompting with Large Language Models
 - b. Divyashanu Swain
- b) Now, can you think and create a new example exemplifying the main conclusion of the paper.
 - a. A new example that exemplifies the main conclusion of the paper could be a case where LLM's are used to answer questions about course registration planning. If you provide the LLM a knowledge graph of all the courses and their prerequisites, the LLM will perform a lot better when recommending courses. Without the knowledge graph, the LLM might only be able to guess which courses require prerequisites, and they might miss some.
- c) Describe how the conclusion is supported in your example.
 - a. The conclusion of the paper is that Graph Neural Prompting can improve the performance of LLM's on certain tasks without having to retrain the LLM. The example I provided shows how providing a knowledge graph of course prerequisites to an LLM could improve the performance on course recommendation tasks without having to do any more training.

Question 2

a) Describe your data preparation, if any, and why or why not. [20 points]

- When I was trying to train classic OCR models, I tried to preprocess the images by converting them to greyscale and increasing contrast to make the text more readable. This only made the text slightly more readable.
- In the end, the only steps I did for data preparation were downloading the data from the google drive, uploading them in batches to ChatGPT, ensuring that the file names followed a consistent format, and that was it. Since ChatGPT not only is capable of OCR but also reasoning, it was able to correct typos and make educated guesses when reading the images.

b) Describe your steps to create a model – pre-trained, your own, manual [30 points]

- I started by researching different OCR libraries within python. I began with TrOCR but got bad results that would be very difficult to work with. I then tried Google Cloud Document API, but I also did not get good results. Finally, I tried the ChatGPT API since I had good results when I uploaded files manually to the browser version, but using the API for images requires a paid subscription. After all these trials I decided to stick with

manually uploading files to ChatGPT5.1 on the browser, specifying that I wanted the output to be in CSV format. ChatGPT was able to generate very accurate outputs compared to the other models I tried, likely because it uses reasoning in addition to just plain OCR. I uploaded them in batches and gave specific instructions for how I wanted the output to be structured. I then took these csv files and added them to my project folder so I could run further analyses on them.

c) Answer the questions from your analyses using the models

a. What are the number of classes and their dates? [10 points]

- There are 26 classes in the provided data (1-27 excluding 23)
- One class, the 23rd class, is missing from this dataset
- The included dates are:
 - Aug: 19th, 21st, 26th, 28th,
 - Sept: 2nd, 4th, 9th, 11th, 16th, 18th, 23rd, 25th, 30th,
 - Oct: 2nd, 7th, 14th, 16th, 21st, 23rd, 28th, 30th
 - Nov: 4th, 11th, 13th, 18th, 20th

b. What is the median class attendance per class [10 points]

- The median attendance is 33 per class

c. What are the dates with lowest and highest attendance? [10 points]

- Lowest attendance: November 20th
- Highest attendance: August 21st

d. Is there a correlation of high attendance with course evaluations dates? When is the attendance highest [10 points]

- Yes there is a correlation between high attendance and course evaluation dates. Oct 7th and Nov 11th had very high attendances (45 and 41 respectively), although November 18th was about average with 34. It is very apparent that the quiz dates led to higher attendance on those days.

d) If you had more time (say a week), what more could you have done to improve performance?

- If I had more time to work on this, I could continue to explore models that I could use in python to streamline the process. Although manually uploading my files to ChatGPT ended up being the most effective and free option, it would be an interesting challenge to try to get decent accuracy with a more bare-bones model, although it likely would not be as accurate as ChatGPT. Additionally, if I had more time I could have generated csv files from multiple different models and compare them for accuracy to hopefully produce the most accurate result. Other than playing around with models, there is not much else I could have done to improve performance